

Yifeng He

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Curriculum Vitae

Research Interests

My research interests lie in AI-driven software security and engineering, as well as in ensuring the security of AI models. I focus on developing novel AI-based techniques for testing software systems, libraries, and applications.

Education

- 2023-present **Ph.D., Computer Science**, *University of California, Davis*.
- 2019-2023 **B.S., Computer Science, Applied Math**, *University of California, Davis*.

Publications

In Conference Proceedings

- 2026 **Yifeng He**, Ziyue Tang, and Hao Chen. Content fuzzing for escaping information cocoons on digital social media. In *Findings of the Association for Computational Linguistics (ACL)*, 2026. [\[link\]](#).
- 2026 Hongxiang Zhang, Yuyang Rong, **Yifeng He**, and Hao Chen. Llamafuzz: Large language model enhanced greybox fuzzing. In *ACM/IEEE International Conference on Automation of Software Test (AST)*, 2026. [\[link\]](#).
- 2025 **Yifeng He**, Luning Yang, Christopher Gonzalo, and Hao Chen. Evaluating program semantics reasoning with type inference in system f. In *Neural Information Processing Systems (NeurIPS)*, 2025. [\[link\]](#).
- 2025 **Yifeng He**, Jicheng Wang, Yuyang Rong, and Hao Chen. FuzzAug: Data augmentation by coverage-guided fuzzing for neural test generation. In *Findings of Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025. [\[link\]](#).
- 2025 **Yifeng He**, Ethan Wang, Yuyang Rong, Zifei Cheng, and Hao Chen. Security of ai agents. In *International Workshop on Responsible AI Engineering (RAIE)*, 2025. [\[link\]](#).
- 2024 Jiabo Huang, Jianyu Zhao, Yuyang Rong, Yiwen Guo, **Yifeng He**, and Hao Chen. Code representation pre-training with complements from program executions. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024. [\[link\]](#).
- 2024 **Yifeng He**, Jiabo Huang, Yuyang Rong, Yiwen Guo, Ethan Wang, and Hao Chen. UniTSyn: A large-scale dataset capable of enhancing the prowess of large language models for program testing. In *International Symposium on Software Testing and Analysis (ISSTA)*, 2024. [\[link\]](#).
- 2023 Jianyu Zhao, Yuyang Rong, Yiwen Guo, **Yifeng He**, and Hao Chen. Understanding programs by exploiting (fuzzing) test cases. In *Findings of the Association for Computational Linguistics (ACL)*, 2023. [\[link\]](#).
- 2022 **Yifeng He**. Big data and deep learning techniques applied in intelligent recommender systems. In *2022 IEEE 4th International Conference on Civil Aviation Safety and Information Technology (ICCSIT)*, pages 1119–1124, 2022. [\[link\]](#).

Preprints

- 2026 Jicheng Wang, **Yifeng He**, Zili Wang, Hanwen Xing, Arkaprava De, and Hao Chen. Contractbench: Can llm agents preserve observation contracts?, 2026. [\[link\]](#).
- 2026 **Yifeng He**, Ethan Wang, Jicheng Wang, Xuanxin Ouyang, and Hao Chen. Code generation by differential test time scaling, 2026. [\[link\]](#).
- 2026 Xiangyi Li, Wenbo Chen, Yimin Liu, Shenghan Zheng, Xiaokun Chen, **Yifeng He**, et al. Skillsbench: Benchmarking how well agent skills work across diverse tasks, 2026. [\[link\]](#).
- 2024 Hongxiang Zhang, **Yifeng He**, and Hao Chen. Steerdiff: Steering towards safe text-to-image diffusion models, 2024. [\[link\]](#).
- 2024 Jicheng Wang, **Yifeng He**, and Hao Chen. Repogenreflex: Enhancing repository-level code completion with verbal reinforcement and retrieval-augmented generation, 2024. [\[link\]](#).

Services

Organizing Committee

2026 The First Workshop on Agent Skills (Agent Skills '26)

Reviewer / Program Committee

2026 ACM Conference on AI and Agentic Systems (CAIS), Main & Demo

2026 Conference on Language Modeling (COLM)

2025,2026 International Conference on Learning Representations (ICLR)

2026 Neural Information Processing Systems (NeurIPS)

2026 International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA)

2025,2026 International Conference on Multimedia (MM)

2025 International Conference on Software Engineering Advances (ICSEA)

2025 Association for Computational Linguistics LLM Security Workshop (ACL-LLMSEC)

Artifact Evaluation Committee

2026 IEEE Symposium on Security and Privacy (S&P)

2025 Conference of the ACM Special Interest Group on Data Communication (SIGCOMM)

Awards

2023 **Citation for Outstanding Performance**, Department of Mathematics, UC Davis

2019-2022 **Dean's Honor List**, College of Letters and Science, UC Davis

Industry Experiences

06/2026- **Applied Scientist Intern**, *Amazon Web Services (AWS)*, Kiro Science.

09/2026

- Research on making agents smarter on analyzing code

04/2021- **Software Development Engineer Intern**, *ByteDance (Toutiao)*, Creator Income Platform.

07/2021

- Used microservice tech to connect parts of the author income settlement business
- Transformed author-relation data architecture design from relational database (SQL) to graph database (Gremlin) to allow better efficiency for the business model
- Refactored the income calculation control process within Python 3

07/2021- **Software Development Engineer Intern**, *ByteDance (Xigua Video)*, Creator Experience Team.

08/2021

- Created a data cleaner with ORM to maintain the size and readability of the online data settlement table
- Created the offline flow of Medium-Length Video Encouragement Project for weekly data calculation
- Built the interface for the frontend of the web and mobile app to display the data visualization